AI for Visualization

4DV805 / 4DT909 - Information Visualization

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Overview



AI4VIS

AI assisting the visualization pipeline.



VIS4AI

Visualization aiding ML interpretability.

Overview



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AI assisting the visualization pipeline.



VIS4AI

Visualization aiding ML interpretability.

AI in Visual Analytics

ML supporting data analytics in the VA process.

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Goals of AI in Visualization

Four main goals:

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Generation

Data to Visual Representation.

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Data to Visual Representation.

Enhancement
Optimizing Visual Representation.

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Predicting and parsing user intent.

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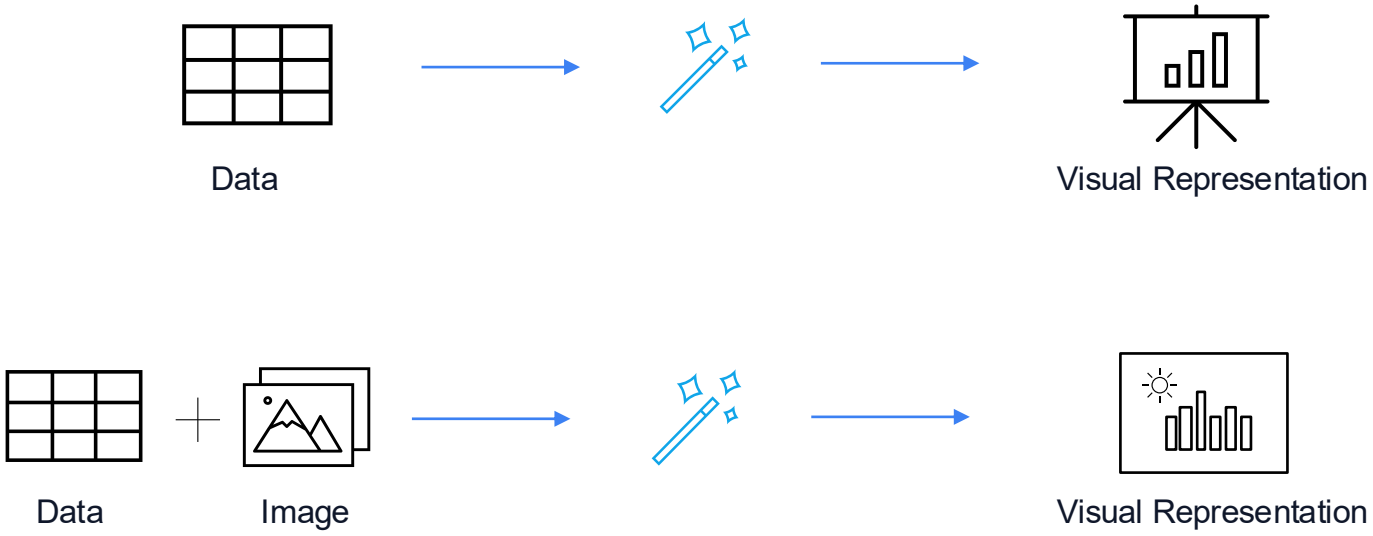
Interaction
Predicting and parsing user intent.

Wu, Aoyu, Yun Wang, Xinhuan Shu, Dominik Moritz, Weiwei Cui, Haidong Zhang, Dongmei Zhang, and Huamin Qu. "Ai4vis: Survey on artificial intelligence approaches for data visualization." IEEE Transactions on Visualization and Computer Graphics 28, no. 12 (2021): 5049-5070.

Visualization Generation

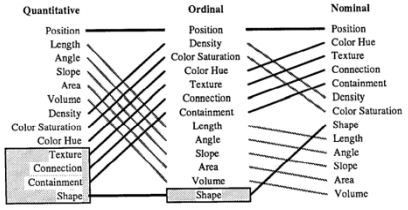


Visualization Generation:



Visualization Generation – Rule Based

APT: Formalized expressiveness and effectiveness criteria.



	Nominal	Ordinal	Quantitative
Size	–	•	•
Saturation	–	•	•
Texture	•	•	
Color	•	•	
Orientation	•		
Shape	•		

Encoding Technique	Primitive Graphical Language
Single-position	Horizontal axis, vertical axis
Apposed-position	Line chart, bar chart, plot chart
Retinal-list	Color, shape, size, saturation, texture, orientation
Map	Road map, topographic map
Connection	Tree, acyclic graph, network
Misc. (angle, contain, . . .)	Pie chart, Venn diagram, . . .

Encoding Technique	Expressiveness Criteria
Single-position	$X \rightarrow Y$ (X is nominal)
Apposed-position	$X \times Y$ (X, Y are not nominal)
Retinal-list	X, or $X \rightarrow Y$ (X is not quantitative)
Map	$L \rightarrow X_1, \ldots$ (L is a location)
Connection	$X \times X$ (X is nominal)
Miscellaneous (angle, contain, . . .)	Generally, $X \times Y$

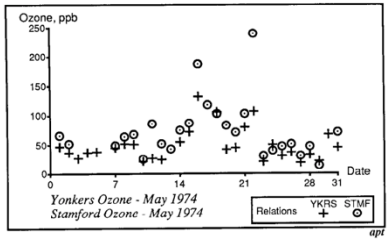


Fig. 26. Example of double-axes composition. The diagram describes a month of ozone measurements for two cities. The reason that a line chart design was not used for these data is that there are missing measurements for some of the days; a line chart is only used for continuous functions.

Mackinlay, Jock. "Automating the design of graphical presentations of relational information." Acm Transactions On Graphics (Tog) 5, no. 2 (1986): 110-141.

Visualization Generation – Rule Based

Show Me: Tableau's core heuristic decision tree.

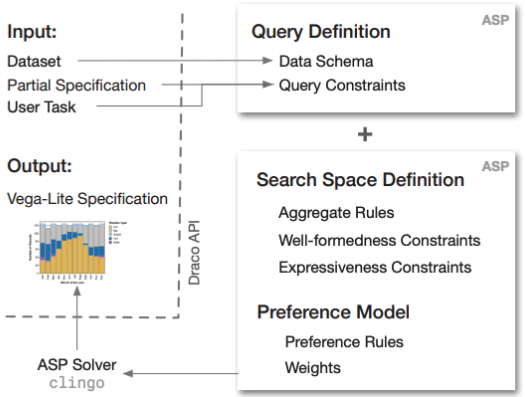
Pane Type		Mark Type	View Type
Field	Field		
C	C	Text	Cross-tab
Qd	C	Bar	Bar view
Qd	Cdate	Line	Line view
Qd	Qd	Shape	Scatter plot
Qi	C	Gantt	Gantt view
Qi	Qd	Line	Line view
Qi	Qi	Shape	Scatter plot

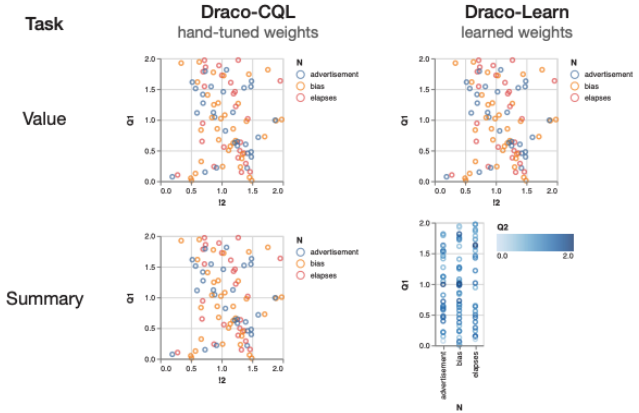


Mackinlay, Jock, Pat Hanrahan, and Chris Stolte. "Show me: Automatic presentation for visual analysis." IEEE transactions on visualization and computer graphics 13, no. 6 (2007): 1137-1144.

Visualization Generation - Supervised ML & Optimization

Draco-Learn: Constraint programming. RankSVM weights derived from human perceptual preference studies.

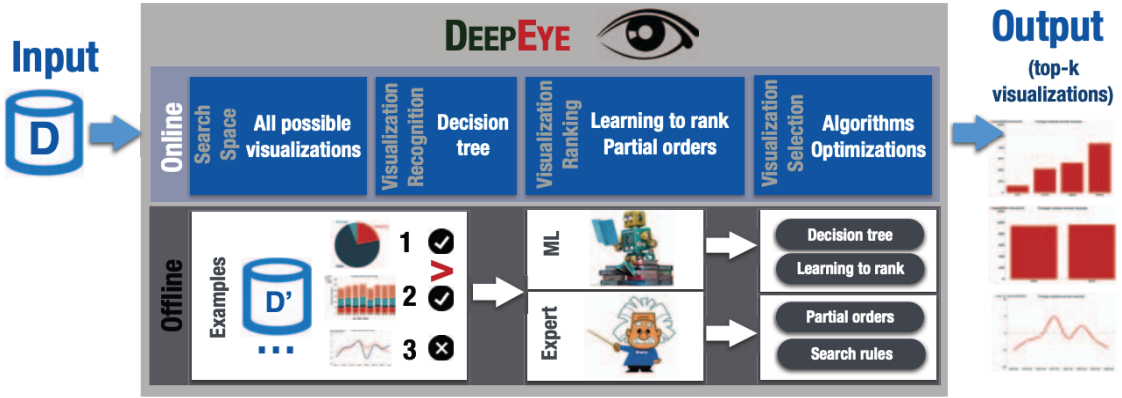




Moritz, Dominik, Chenglong Wang, Greg L. Nelson, Halden Lin, Adam M. Smith, Bill Howe, and Jeffrey Heer. "Formalizing visualization design knowledge as constraints: Actionable and extensible models in draco." IEEE transactions on visualization and computer graphics 25, no. 1 (2018): 438-448.

Visualization Generation - Supervised ML & Optimization

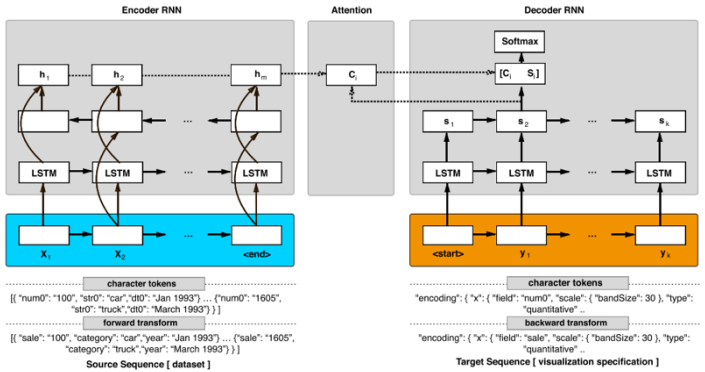
DeepEye: Binary classification models scoring and ranking the viability of generated charts.



Luo, Yuyu, Xuedi Qin, Nan Tang, and Guoliang Li. "Deepeye: Towards automatic data visualization." In 2018 IEEE 34th international conference on data engineering (ICDE), pp. 101-112. IEEE, 2018.

Visualization Generation - Supervised ML & Optimization

Data2Vis: Neural machine translation. JSON data directly mapped to Vega-Lite via LSTMs.

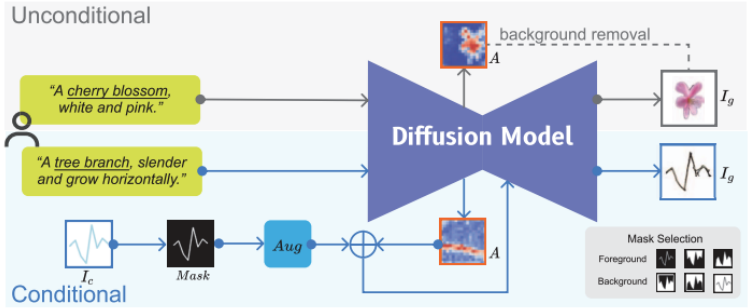


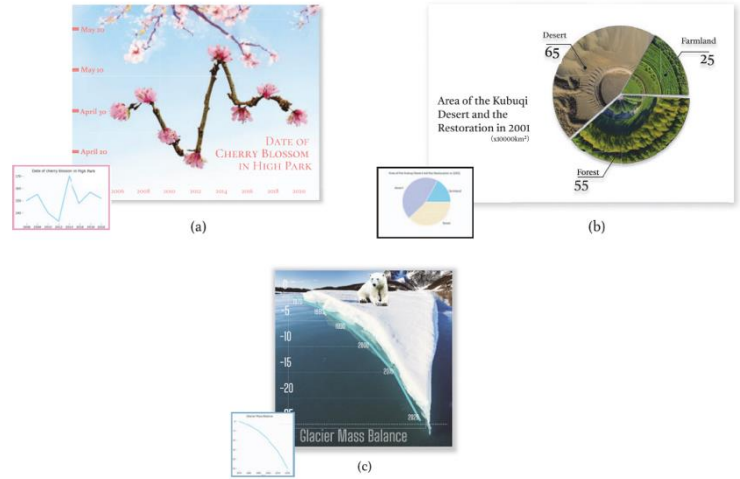


Dibia, Victor, and Çağatay Demiralp. "Data2vis: Automatic generation of data visualizations using sequence-to-sequence recurrent neural networks." IEEE computer graphics and applications 39, no. 5 (2019): 33-46.

Visualization Generation - Generative Models

Let the Chart Spark: Diffusion-based embellished chart generation.

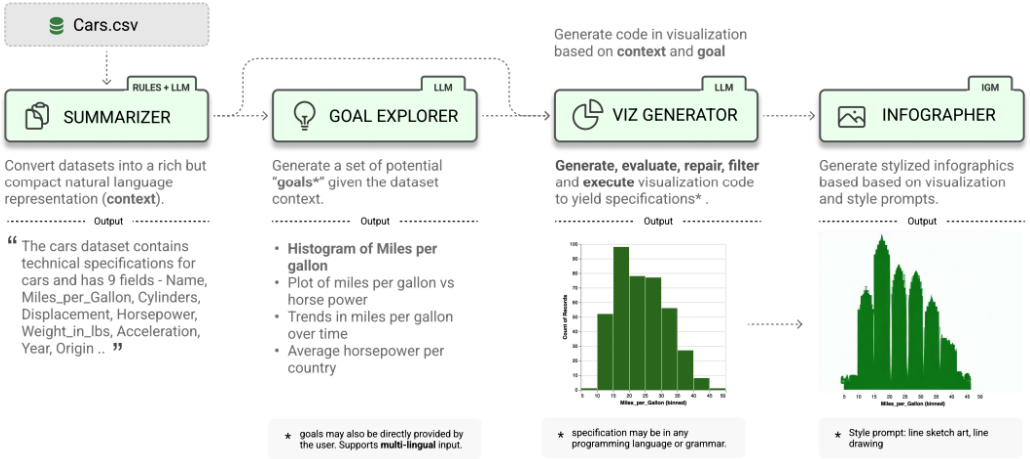




Xiao, Shishi, Suizi Huang, Yue Lin, Yilin Ye, and Wei Zeng. "Let the chart spark: Embedding semantic context into chart with text-to-image generative model." IEEE Transactions on Visualization and Computer Graphics 30, no. 1 (2023): 284-294.

Visualization Generation - Generative Models

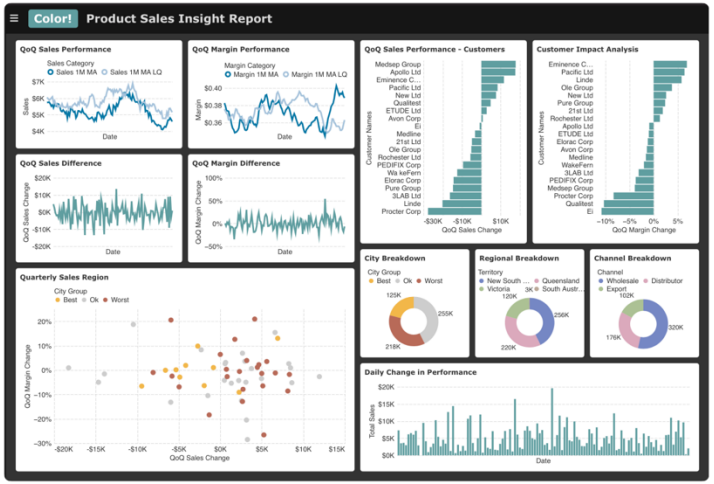
LIDA: LLM-based orchestrated pipeline that goes from data profiling from generation.



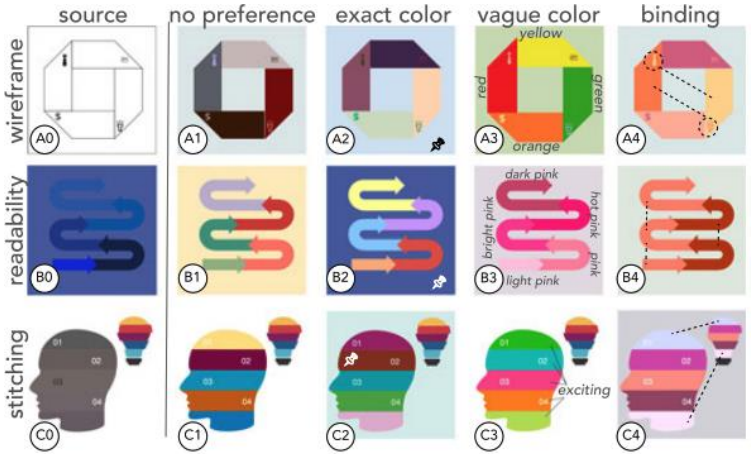
Dibia, Victor. "LIDA: A tool for automatic generation of grammar-agnostic visualizations and infographics using large language models." In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations), pp. 113-126. 2023.

Visualization Enhancement – Color Recommendation

MVcolor



Infocolorizer

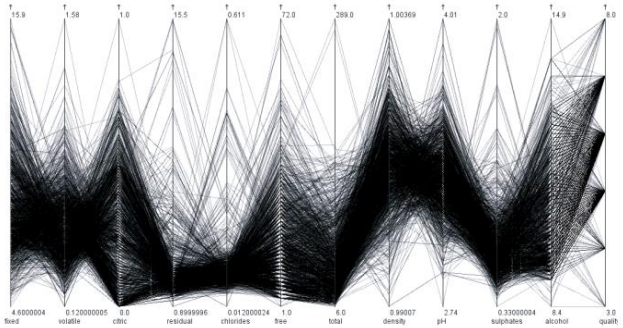


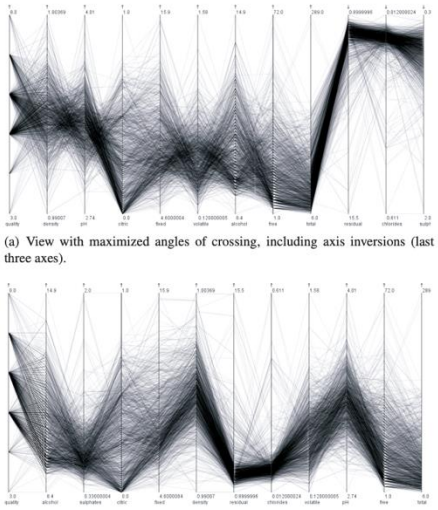
Lin, Yun-Rou, Fu-Yin Cherng, Yi-Chieh Lee, Zhu-Ying Tian, and Wen-Chieh Lin. "Navigating Color Constraints in Multi-View Visualizations with MVcolor." In 2025 IEEE 18th Pacific Visualization Conference (PacificVis), pp. 138-148. IEEE, 2025.

Yuan, Lin-Ping, Ziqi Zhou, Jian Zhao, Yiqiu Guo, Fan Du, and Huamin Qu. "Infocolorizer: Interactive recommendation of color palettes for infographics." IEEE Transactions on Visualization and Computer Graphics 28, no. 12 (2021): 4252-4266.

Visualization Enhancement – Layouts

Parallel Coordinate Axes Ordering





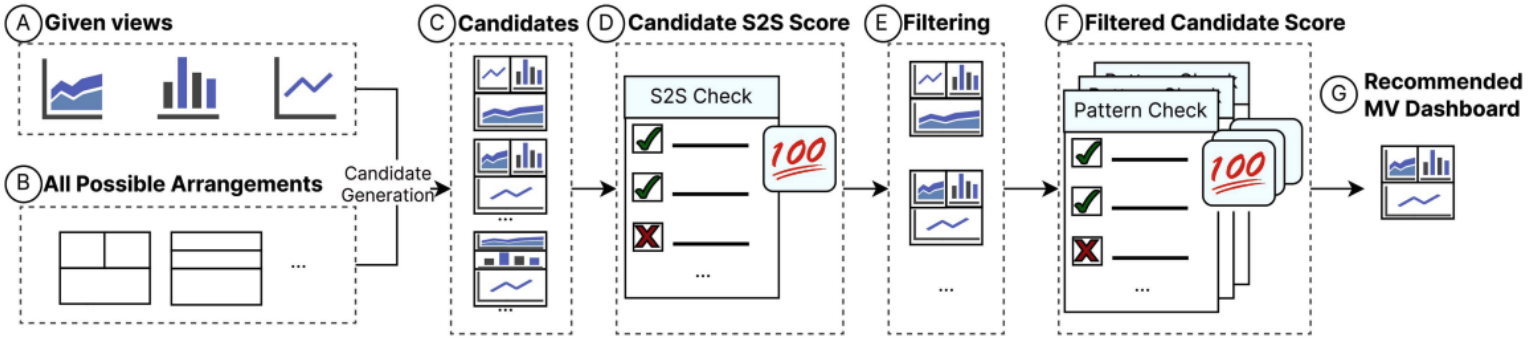
(a) View with maximized angles of crossing, including axis inversions (last three axes).

(b) Minimizing the number of crossings.

Dasgupta, Aritra, and Robert Kosara. "Pargnostics: Screen-space metrics for parallel coordinates." IEEE Transactions on Visualization and Computer Graphics 16, no. 6 (2010): 1017-1026.

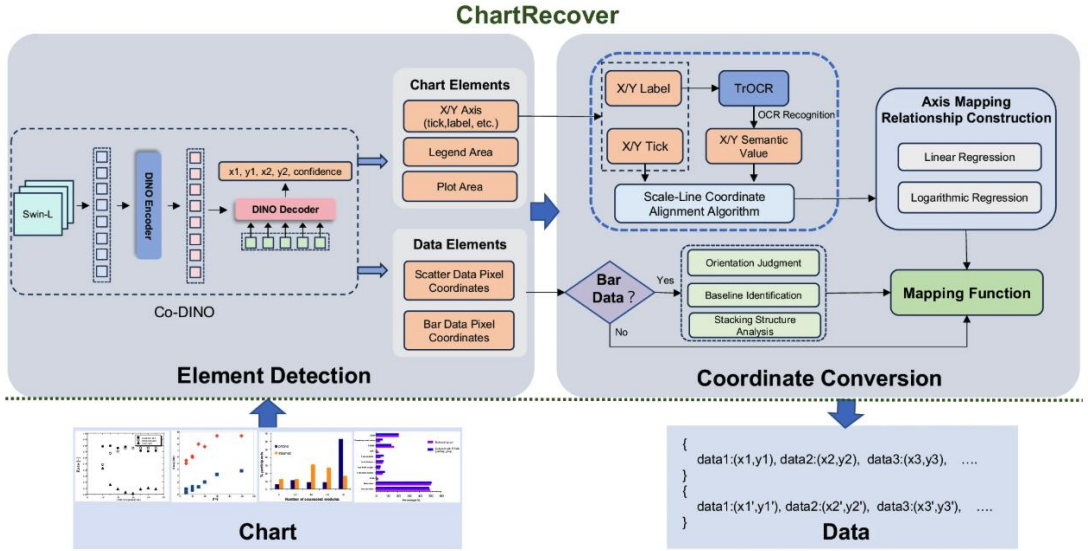
Visualization Enhancement – Layouts

Dashboard Layout



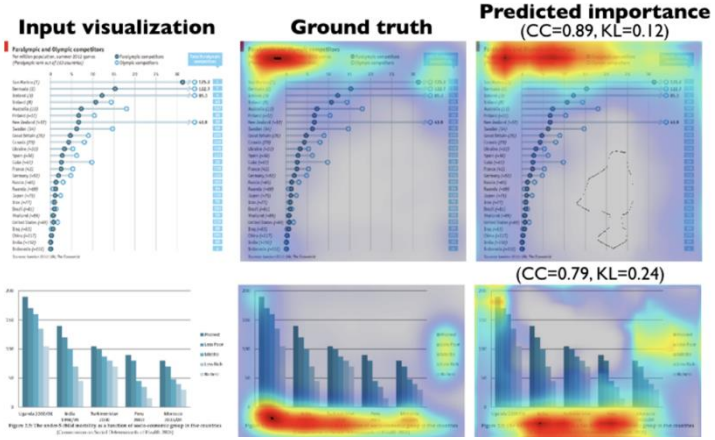
Lin, Yanna, Haotian Li, Aoyu Wu, Yong Wang, and Huamin Qu. "DMiner: Dashboard design mining and recommendation." IEEE Transactions on Visualization and Computer Graphics 30, no. 7 (2023): 4108-4121.

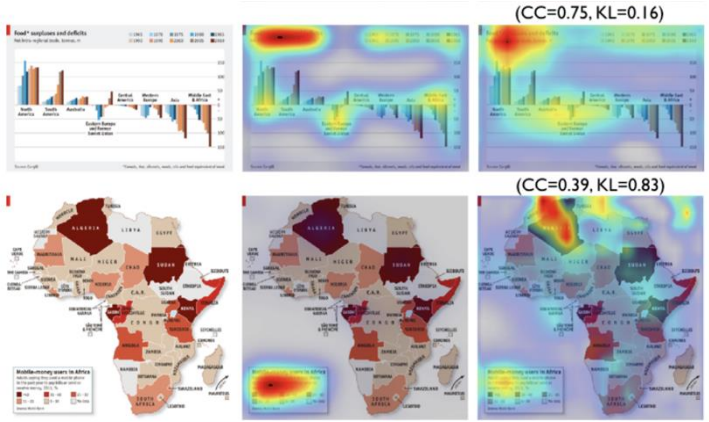
Visualization Analysis – Data Extraction



Yuan, Yang, Sihan Liang, Ji Zhang, Yangang Wang, and Zongguo Wang. "A deep learning framework for scientific chart data extraction and reconstruction." Communications Engineering (2026).

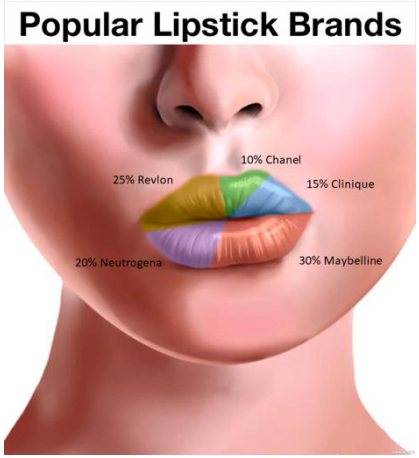
Visualization Analysis – ROI Identification





Bylinskii, Zoya, Nam Wook Kim, Peter O'Donovan, Sami Alsheikh, Spandan Madan, Hanspeter Pfister, Fredo Durand, Bryan Russell, and Aaron Hertzmann. "Learning visual importance for graphic designs and data visualizations." In Proceedings of the 30th Annual ACM symposium on user interface software and technology, pp. 57-69. 2017.

Visualization Analysis – Distortion Estimation

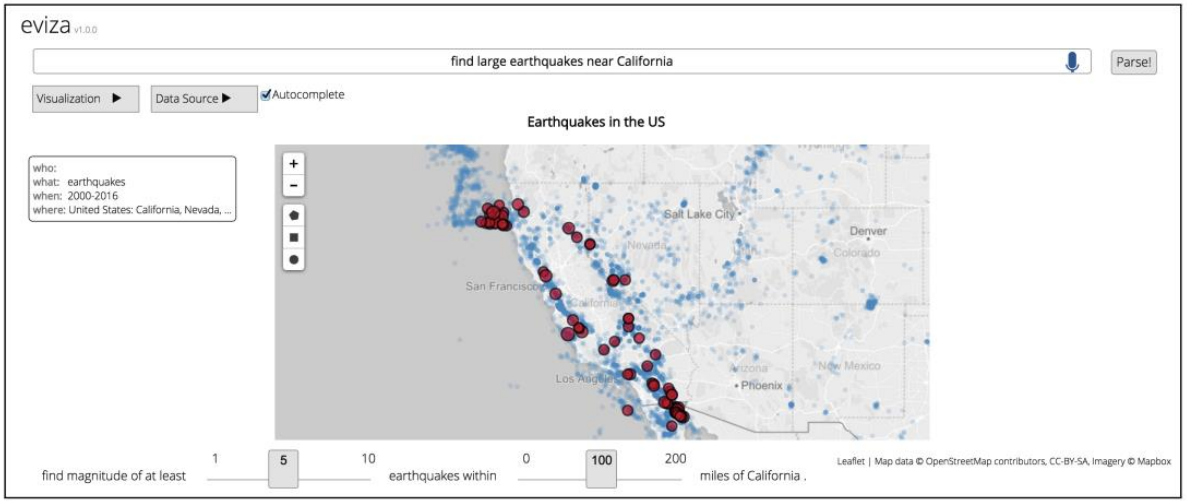




Coelho, Darius, and Klaus Mueller. "Infomages: Embedding data into thematic images." In Computer Graphics Forum, vol. 39, no. 3, pp. 593-606. 2020.

Visualization Interaction

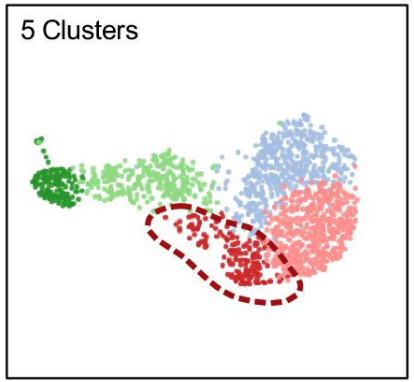
NLI for Interaction

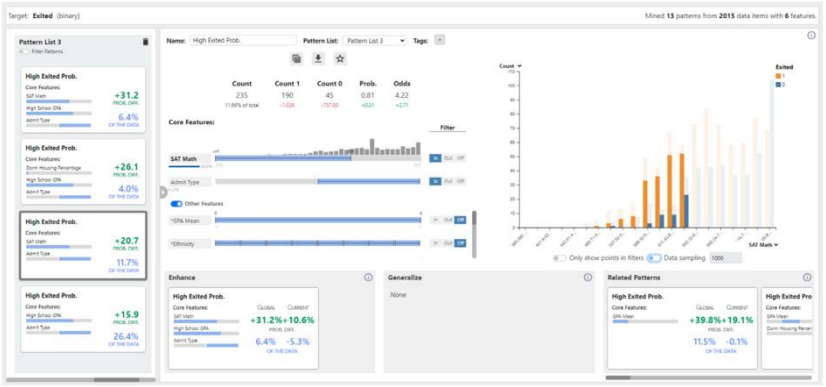


Setlur, Vidya, Sarah E. Battersby, Melanie Tory, Rich Gossweiler, and Angel X. Chang. "Eviza: A natural language interface for visual analysis." In Proceedings of the 29th annual symposium on user interface software and technology, pp. 365-377. 2016.

Visualization Interaction

Sub-Space Identification

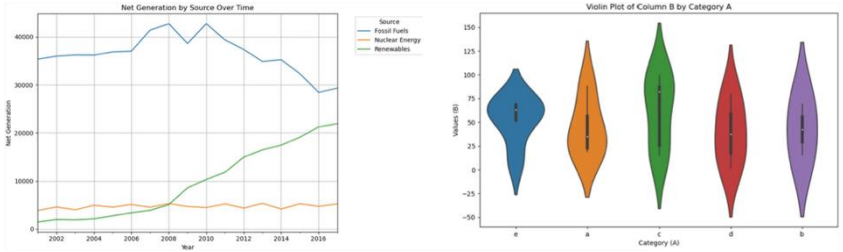


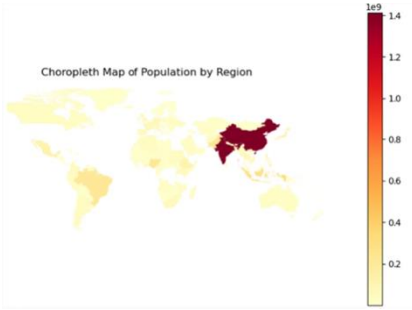


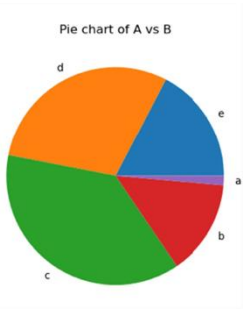
Coelho, Darius, Eric Papenhausen, and Klaus Mueller. "Evolutionary design of a visual analytics interface to study predictive patterns in high dimensional data." Visual Informatics (2025): 100303.

LLMs, VLLMs, and MLLMs

LLM Generated Charts







Khan, Saadiq Rauf, Vinit Chandak, and Sougata Mukherjea. "Evaluating LLMs for visualization generation and understanding." Discover Data 3, no. 1 (2025): 15.

Writing Good Prompts

Visualize this data.

Writing Good Prompts

Visualize this data.



Visualize this data with a bar chart.

Writing Good Prompts

Visualize this data.



Visualize this data with a bar chart.



Write a python script to visualize this data with a bar chart.x

Writing Good Prompts

Visualize this data.



Visualize this data with a bar chart.



Write a python script to visualize this data with a bar chart.



Write a python script to visualize this data with a bar chart. Use [xyz] principals.

Pitfalls

Data Alteration

Generating bins or applying smoothing incorrectly.

Range Misalignment

Selecting axis ranges that mislead

Hallucinated Patterns

Inventing visual structure where it doesn’t statistically exist.

Semantic Gaps

Poor labeling and missing context.

Break the Problem Down

Generic Prompt

I am about to show you an image and ask you a multiple-choice question about that image. Select the BEST answer based only on the chart and not external knowledge.

Charts-of-Thought Prompt

I am about to show you a graph and ask you a multiple-choice question about that graph.
Task 1: Data Extraction and Table Creation: First, explicitly list ALL numerical values you can identify on both axes, then create a structured table using markdown syntax that includes ALL data points you identified above with appropriate column headers with units.
Task 2: Sort the data: Sort the data in descending order by the numerical values.
Task 3: Data Verification and Error Handling: Double-check if your table matches ALL elements in the graph by comparing each value in your table with the graph and updating your table with correct values; verify the sorting is correct, and before proceeding, confirm all corrections have been made and use ONLY the corrected data for analysis.
Task 4: Question Analysis: Using ONLY the verified data in your table, compare EACH value individually with the reference value; for "less than" comparisons, mark ALL values that are even slightly below the reference; for "greater than" comparisons mark ALL values that are even slightly above the reference, and show each comparison on a new line.
Let's solve this step by step.

Das, Amit Kumar, Mohammad Tarun, and Klaus Mueller. "Charts-of-thought: Enhancing llm visualization literacy through structured data extraction." IEEE Transactions on Visualization and Computer Graphics (2025).

Generate Code/Specifications

Imperative
(Matplotlib / D3)

```
fig, ax = plt.subplots()
 ax.bar(x - width/2, y1, width)
 ax.bar(x + width/2, y2, width)
 ax.set_xticks(x, labels)
 fig.tight_layout()
// Manual coordinate handling...
```

Declarative
(Vega-Lite)

```
"mark": "bar",
 "encoding": {
   "x": {"field": "category"},
   "y": {"field": "value", "aggregate": "sum"},
   "color": {"field": "type"}
 }
// Deterministic rendering.
```

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```

Requires spatial math and state management. Higher failure rate.

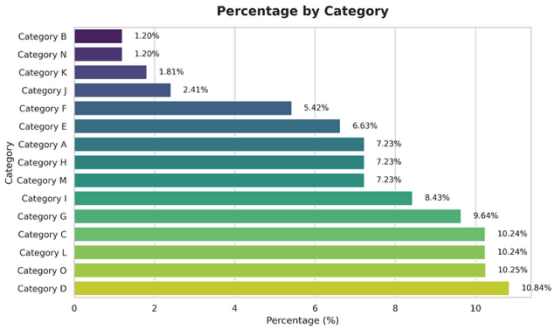
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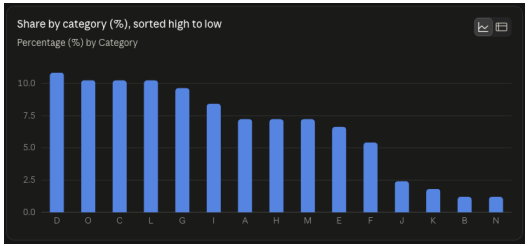
Separates specification from execution. Schema-validatable.

Example

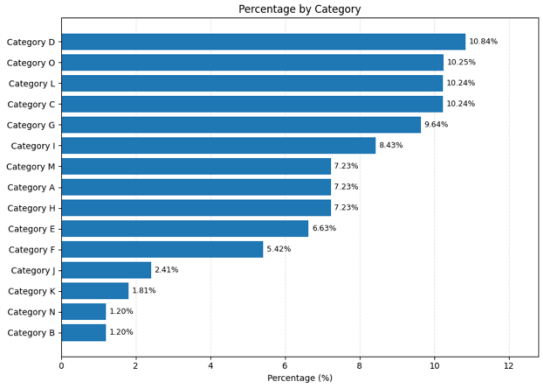
Gemini



Claude

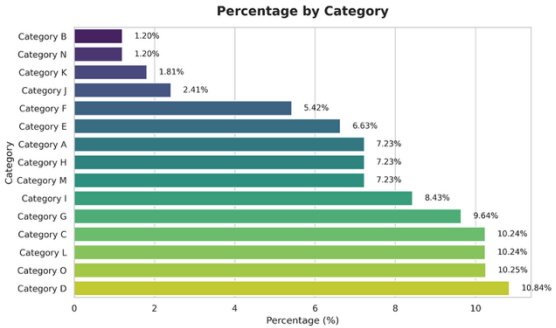


ChatGPT

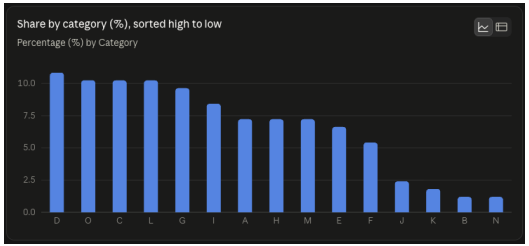


Example

Gemini

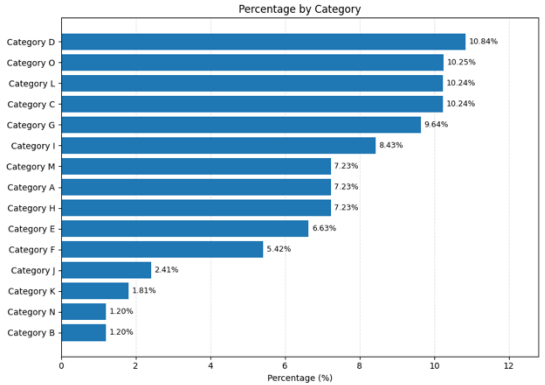


Claude



I can also make a pie or donut chart, or keep the original category order, if you'd prefer.

ChatGPT



A horizontal bar chart is a good choice here because it makes it easy to compare all 15 categories. A pie chart would also be possible, but the similar-sized slices would make precise comparisons more difficult.

Evaluate Results

Visual Design & Perceptual Accuracy

Validating legibility, overlap, and density.

Statistical Faithfulness

Checking for zero hallucination in metrics.

Code Compilation

Ensuring syntax validity and execution success.

Discussion

If AI can/will do everything,
why visualize data?

Why visualize data?

Trust

Uncertainty

Trade offs

Empty Spaces